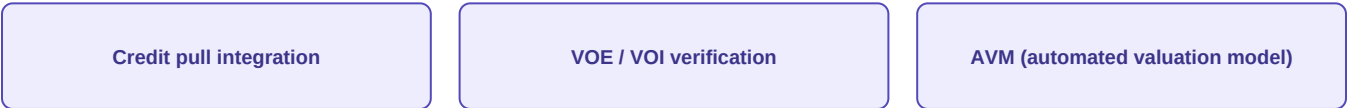
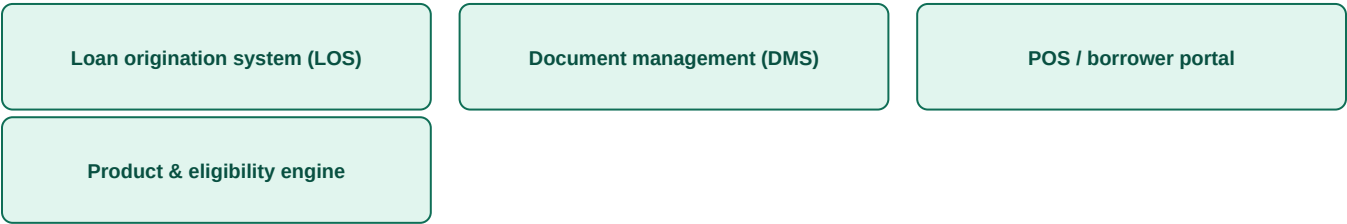


Phase 1: Data foundation



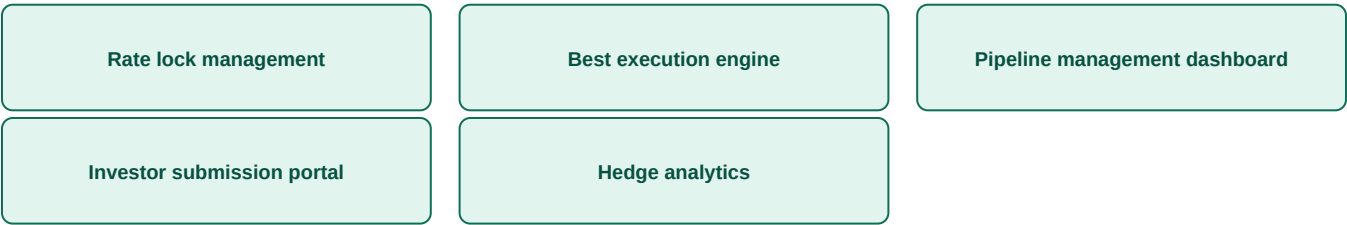
Phase 2: Core origination



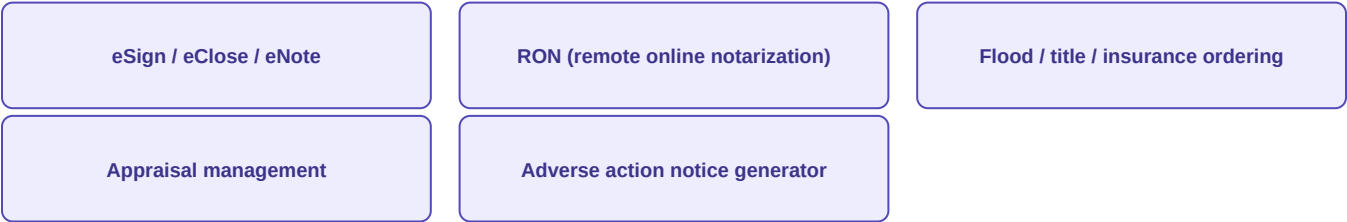
Phase 3: Pricing & compliance



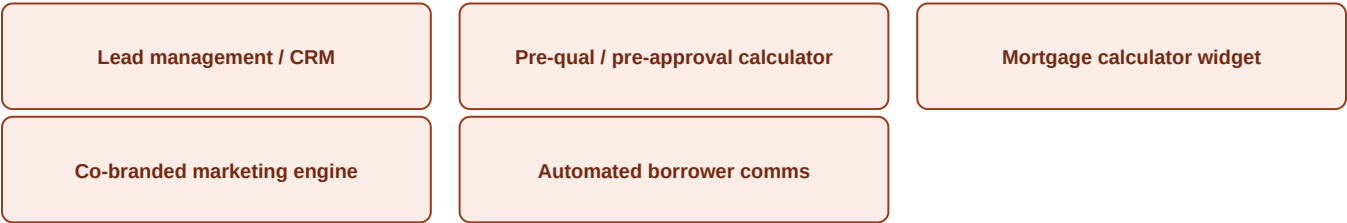
Phase 4: Rate lock & secondary market



Phase 5: Closing & digital execution



Phase 6: Borrower experience & leads



Phase 7: Compliance reporting & licensing



Quick pricer vs. pricing engine

Dimension	Quick pricer	Pricing engine
Purpose	Ballpark estimate for early conversations	Lock-ready, precise rate quote
Who uses it	Borrowers, realtors, LOs on phone	Loan officers, lock desk, processors
Inputs required	3-5 fields: loan amt, score, property	15-30+ fields: full borrower profile
Rate data source	Simplified internal rate table	Live investor rate sheets (10-50+)
LLPA adjustments	None or simplified buckets	Full grid: FICO x LTV x property x occ
Accuracy	+/- 0.25-0.50% from final rate	Exact to basis point — lockable
Speed	Instant (< 1 second)	2-5 seconds (multiple investor APIs)
Investor coverage	0 — uses generic/internal rates	10-50+ investors, real-time
Can you lock?	No — estimate only	Yes — direct lock with investor
Comp / margin	Not included	Built-in: LO, branch, corporate layers
Integration	Standalone widget / API	Deep LOS, lock desk, secondary
Pricing model	Free / low-cost add-on	\$500-2,000+/mo per agency

Phase 1: Data foundation

Build first — everything depends on these

Dependencies:

None — this is the foundation layer.

Feeds into:

Phase 2 (Core origination) — LOS and eligibility engine consume credit, income, and property data.

Modules in this phase:

Credit pull integration

Tri-merge credit reports from Equifax, Experian, TransUnion. Provides FICO scores, trade lines, and liability data needed by every downstream module.

VOE / VOI verification

Verify employment and income via The Work Number, Plaid, or direct employer verification. Feeds into AUS and eligibility.

AVM (automated valuation model)

Instant property value estimates from comparable sales data, tax records, and statistical models. Used by eligibility and pricing engines.

Phase 2: Core origination

The backbone — loan lifecycle management

Dependencies:

Phase 1 — needs credit, income, and property data to populate loan files and determine eligibility.

Feeds into:

Phase 3 (Pricing & compliance) — eligibility results and borrower profiles drive pricing and compliance checks.

Modules in this phase:

Loan origination system (LOS)

End-to-end loan lifecycle from application intake to closing. Central database for all loan data. Every other module reads from or writes to the LOS.

Document management (DMS)

Store, organize, OCR, and retrieve loan documents. Integrates with LOS for document checklists and audit trails.

POS / borrower portal

Online application and document upload for borrowers. Front-end that feeds data into the LOS. Reduces manual data entry.

Product & eligibility engine

Matches borrower profile (credit, income, property) to eligible loan programs (Conventional, FHA, VA, USDA, Jumbo). Filters before pricing.

Phase 3: Pricing & compliance

Needs eligibility data + borrower profile from Phase 2

Dependencies:

Phase 2 — pricing engine needs eligible products and full borrower profile. Compliance engine validates data captured in the LOS.

Feeds into:

Phase 4 (Rate lock & secondary) — pricing output enables rate locking and investor comparison.

Modules in this phase:

Pricing engine

Connects to 10-50+ investor rate sheets. Applies full LLPA grid (FICO x LTV x property type x occupancy x loan purpose). Returns lock-ready, basis-point-accurate quotes.

Quick pricer

Simplified front-end tool using 3-5 inputs for ballpark rate estimates. Used in early borrower conversations. Not lockable.

Compliance engine

Real-time TRID, RESPA, HMDA, and NY state fair-lending checks. Validates every loan action against regulatory rules. Critical for NY agencies.

Fee sheet / closing cost calculator

Generates Loan Estimate and Closing Disclosure-ready fee breakdowns by county, lender, and loan type.

AUS integration

Sends loan files to Desktop Underwriter (DU) or Loan Prospector (LP) for automated underwriting decisions.

QC / pre-close audit

Automated quality control checks before closing. Catches errors, missing docs, and compliance gaps.

Phase 4: Rate lock & secondary market

Needs pricing engine output to function

Dependencies:

Phase 3 — rate locks require pricing engine quotes. Best execution compares across investors connected in the pricing engine.

Feeds into:

Phase 5 (Closing) — locked, compliant loans flow into closing and document execution.

Modules in this phase:

Rate lock management

Lock, extend, re-lock, and float-down rates with full audit trail. Tracks lock expiration and manages lock desk workflow.

Best execution engine

Compares pricing across all connected investors to find the most profitable execution for each loan. Factors in SRP, margin, and investor overlays.

Pipeline management dashboard

Real-time view of every loan in flight — status, bottlenecks, SLA tracking, and aging alerts.

Investor submission portal

Package and deliver complete loan files to investors for purchase review. Manages conditions and stipulations.

Hedge analytics

Manages interest rate risk on the pipeline. Models pull-through rates, coverage ratios, and P&L impact of rate movements.

Phase 5: Closing & digital execution

Needs locked loan + compliant docs from Phases 3-4

Dependencies:

Phases 3-4 — closing requires locked rates, compliant documents, and completed underwriting.

Feeds into:

Phase 6 (Borrower experience) — a working pipeline gives context for lead nurturing and borrower communication.

Modules in this phase:

eSign / eClose / eNote

Digital signing of loan documents, electronic closing, and creation of tamper-evident electronic promissory notes.

RON (remote online notarization)

Video-based remote notarization compliant with NY state RON laws. Enables fully digital closings.

Flood / title / insurance ordering

Automated ordering of flood certifications, title searches, and insurance policies from third-party vendors.

Appraisal management

Order, track, and review property appraisals. Manages appraiser assignment, scheduling, and report delivery.

Adverse action notice generator

Auto-generates denial letters with correct ECOA reason codes when a loan is declined.

Phase 6: Borrower experience & leads

Layer on top of working origination pipeline

Dependencies:

Phases 1-5 — these tools feed into and sit on top of the origination pipeline. Without a working pipeline, leads have nowhere to go.

Feeds into:

Phase 7 (Compliance reporting) — lead and loan data flows into regulatory reporting.

Modules in this phase:

Lead management / CRM

Track, score, and nurture borrower leads from initial contact through conversion. Integrates with POS for seamless handoff.

Pre-qual / pre-approval calculator

Instant affordability estimates for borrowers based on income, debt, and target area. Feeds qualified leads into the LOS.

Mortgage calculator widget

Embeddable payment calculator for agency websites. Captures leads and provides instant monthly payment estimates.

Co-branded marketing engine

Generate rate sheets, flyers, and marketing materials co-branded with realtor partners.

Automated borrower comms

Automated status updates, document requests, and milestone alerts via email, SMS, and push notifications.

Phase 7: Compliance reporting & licensing

Needs historical loan data from all prior phases

Dependencies:

Phases 1-6 — HMDA reporting requires complete loan data including demographics, geography, and outcomes from closed loans.

Feeds into:

Phase 8 (Analytics) — compliance data enriches analytics and performance tracking.

Modules in this phase:

HMDA reporting

Collect and submit Home Mortgage Disclosure Act data. Aggregates loan-level demographic and geographic data for federal reporting.

NMLS / licensing tracker

Track loan officer licenses, renewal dates, continuing education requirements, and state-specific compliance across all active states.

Phase 8: Analytics & optimization

Only meaningful with real loan volume flowing through

Dependencies:

All prior phases — analytics require real loan volume and complete data across the entire pipeline.

Feeds into:

Feeds back into all phases — insights drive process improvements, pricing adjustments, and staffing decisions.

Modules in this phase:

LO performance dashboard

Track loan officer volume, conversion rates, speed-to-close, and revenue per loan.

Margin & profitability analytics

Per-loan and per-LO revenue and cost breakdown. Identifies margin leaks and optimization opportunities.

Turn time / SLA tracking

Measure processing speed at each stage against service-level targets. Identifies bottlenecks.

Pipeline reporting

Funnel analysis from lead to funded loan. Tracks fallout rates and conversion at each stage.

Investor comparison analytics

Compare pricing, turn times, and kick-out rates across all connected investors.